Forest ReCom: Reversible Recombination via Spanning Forest Sampling

G.9 — Ensemble Methods / Distributional Correctness

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Abstract

Standard ReCom [DeFord et al., 2021] samples spanning tree cuts uniformly at random, accepting every valid balanced cut without correction. This biases the chain toward plans reachable by many spanning tree cuts—those whose district boundaries lie in geometrically accessible regions of the census-tract graph. Forest ReCom [McCartan and Imai, 2020] corrects this bias using a Metropolis-Hastings ratio: given a proposal generated from a forward spanning tree T_{fwd} with c_{fwd} valid balanced cuts, the step is accepted with probability $\min(1, c_{\text{fwd}}/c_{\text{rev}})$, where c_{rev} is the count of valid cuts in an independent reverse spanning tree T_{rev} drawn from the proposed district pair. We implement ForestReCom in `redist-ensemble` as `ForestRecomChain` and wire it into the compositor as `-search forest-recom`. The chain satisfies *expected detailed balance* with the uniform distribution over valid plans as its stationary target: the ratio $c_{\text{fwd}}/c_{\text{rev}}$ is an unbiased estimator of the true transition probability ratio. This is a weaker guarantee than the exact calibration of sequential Monte Carlo [Della-Libera, 2026b] (G.7) but a stronger guarantee than standard ReCom (unknown stationary distribution). On North Carolina ($k = 14$), Wisconsin ($k = 8$), and Texas ($k = 38$), ForestReCom achieves acceptance rates of approximately 40–60% (versus $\approx 80\%$ for standard ReCom) at roughly $2\times$ the per-step computational cost. Verification on a small synthetic graph (4-node path, $k = 2$, 10,000 steps) confirms convergence to the uniform plan distribution; a chi-squared test rejects standard ReCom and accepts ForestReCom at $\alpha = 0.05$.

Contents

1. Introduction

1.1. The Distributional Problem in Standard ReCom

Markov chain redistricting has two distinct goals that are often conflated. The first is *optimisation*: finding the best plan by some criterion (edge cut, compactness, partisan fairness). The second is *sampling*: drawing plans from a target distribution so that ensemble statistics reflect genuine distributional properties of the plan space.

Standard ReCom [DeFord et al., 2021] is widely used for both goals. At each step, it selects an adjacent pair of districts, merges them, samples a random spanning tree of the merged region via Wilson's algorithm, and cuts the tree at a balanced edge. Every valid balanced cut is accepted; the chain is never rejected.

This 100% acceptance rate is computationally appealing, but it creates a distributional bias. Plans reachable by many spanning tree cuts—those whose district boundaries happen to lie in highly-connected regions of the census-tract graph—are visited more often than plans reachable by few cuts. The stationary distribution of standard ReCom is therefore *not* the uniform distribution over valid plans; it is a distribution weighted by the number of spanning tree cuts that produce each plan. DeFord, Duchin, and Solomon [DeFord et al., 2021] acknowledge this but note that characterising the bias analytically is extremely difficult for real census graphs.

For optimisation, this bias is harmless or even beneficial—overweighting geometrically accessible plans can guide the chain toward structurally better solutions. But for ensemble analysis—where the goal is to characterise the full space of valid plans and answer questions like "what fraction of valid plans produce 7 Democratic seats in NC?"—the bias is problematic. Ensemble statistics derived from a biased chain may not represent the space of all valid plans.

1.2. Forest ReCom as the Fix

McCartan and Imai [McCartan and Imai, 2020] introduce ForestReCom to correct this bias. The key idea is to apply a Metropolis-Hastings correction to the ReCom proposal: after generating the proposal from the forward spanning tree T_{fwd}, draw an independent reverse spanning tree T_{rev} from the *proposed* district pair, count its valid balanced cuts, and accept the proposal with probability

$$\alpha = \min\left(1,\ \frac{c_{\text{fwd}}}{c_{\text{rev}}}\right),$$

where c_{fwd} is the number of valid cuts in T_{fwd} and c_{rev} is the number of valid cuts in T_{rev}. If $c_{\text{rev}} = 0$, the proposal is rejected with probability 1.

The intuition is direct: if the forward tree has many more balanced cuts than the reverse tree, the proposed plan is "more accessible" from the current plan than vice versa; the Metropolis-Hastings ratio downweights this asymmetry. Over many steps, the chain converges toward the uniform distribution over valid plans.

ForestReCom does not achieve *exact* detailed balance in the per-step sense because T_{rev} is a single spanning tree sample, not the expectation over all spanning trees. The ratio $c_{\text{fwd}}/c_{\text{rev}}$ is an unbiased estimator of the true transition probability ratio $Q(\pi \to \pi')/Q(\pi' \to \pi)$ only in expectation. We characterise this precisely in Section 3 and follow McCartan and Imai [McCartan and Imai, 2020] §3.2 in calling the resulting guarantee *expected detailed balance*.

1.3. Three Contributions

1. **Implementation.** We provide the complete specification and implementation of ForestReCom as `ForestRecomChain` in `redist-ensemble`, including the deterministic forward/reverse seed schedule, pair selection with reselection, and the acceptance criterion (Section 3).

2. **Comparison.** We compare ForestReCom against standard ReCom on NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$) across acceptance rate, edge-cut distribution, partisan outcomes, and runtime (Section 4).

3. **Distributional correctness analysis.** We verify convergence to the uniform plan distribution on a small synthetic graph where the exact uniform distribution is computable, and use a chi-squared test to quantify the improvement over standard ReCom (Section 5).

1.4. Paper Structure

Section 2 reviews standard ReCom, the spanning-tree selection bias, and Wilson's UST algorithm. Section 3 gives the formal ForestReCom algorithm and proves expected detailed balance. Section 4 presents empirical comparisons on real states. Section 5 establishes distributional correctness on a synthetic benchmark. Section 6 concludes with usage guidance and open questions.

2. Background

2.1. Standard ReCom

Let $G = (V, E)$ be the census-tract adjacency graph for a state, with $|V| = n$ tracts. A redistricting plan $\pi : V \to [k]$ is valid if every district $D_i = \pi^{-1}(i)$ is connected in G and population-balanced: $|P(D_i) - P_{\text{ideal}}| \leq \epsilon \cdot P_{\text{ideal}}$, where $P_{\text{ideal}} = \sum_v \text{pop}(v)/k$ and $\epsilon = 0.005$.

Standard ReCom [DeFord et al., 2021] defines a Markov chain on valid plans $\pi \in \Pi(G, k)$ as follows. At each step:

1. Select an adjacent district pair (d_i, d_j) uniformly at random from all pairs sharing a graph edge.
2. Let $R = D_i \cup D_j$ (the merged region).
3. Sample a random spanning tree $T \sim \text{UST}(G[R])$.
4. Let $C(T)$ be the set of edges $e \in T$ such that removing e produces a balanced (A, B) bipartition of R.
5. If $C(T) = \emptyset$: reject (stay at π).
6. Otherwise: sample $e^* \sim \text{Uniform}(C(T))$; let π' be the plan with $D'_i = A$, $D'_j = B$.
7. Accept: set $\pi \leftarrow \pi'$.

Steps 5–7 are unconditional: every non-empty balanced cut is accepted. The chain has no rejection step beyond the empty-cut failure in step 5.

2.2. The Balanced-Cut Selection Bias

The transition probability from π to π' under standard ReCom is proportional to the expected number of spanning trees of $G[R]$ that contain a balanced cut producing (A, B):

$$Q_{\text{std}}(\pi \to \pi') \ \propto\ \mathbf{E}_{T\sim \text{UST}(G[R])}\Big[\mathbf{1}[T \text{ cuts to } (A, B)]\Big].$$

For the chain to be reversible with respect to the uniform distribution μ_{unif} over $\Pi(G, k)$, we need detailed balance: $\mu_{\text{unif}}(\pi) \cdot Q(\pi \to \pi') = \mu_{\text{unif}}(\pi') \cdot Q(\pi' \to \pi)$. Since $\mu_{\text{unif}}(\pi) = \mu_{\text{unif}}(\pi')$, this requires $Q(\pi \to \pi') = Q(\pi' \to \pi)$—that is, the expected number of trees containing a (A, B) cut from π's side must equal the expected number containing the same cut from π''s side.

In general, this is false. The merged region $R = D_i \cup D_j$ is the same graph regardless of direction, so one might expect the forward and reverse transition probabilities to be equal. However, the pair-selection probabilities differ: the probability of selecting the pair (d_i, d_j) may differ between π and π' because the set of adjacent district pairs depends on the plan. Moreover, even fixing the pair, the number of valid balanced cuts per spanning tree is the same from both sides only if the spanning tree distribution is symmetric with respect to the (A, B) labelling, which need not hold for general balanced cuts.

Remark 1 (Stationarity of standard ReCom)**.** *The stationary distribution of standard ReCom is not the uniform distribution over valid plans $\Pi(G, k)$. It weights plans by their number of spanning tree "entry paths"—plans reachable by many distinct spanning tree cuts are overrepresented. DeFord et al. [DeFord et al., 2021] do not claim that standard ReCom targets the uniform distribution; they use it as a heuristic sampler whose output is interpreted relative to the chain's own distribution. For ensemble analysis requiring a specific target distribution, a corrected chain is needed.*

2.3. Wilson's Uniform Spanning Tree Algorithm

Wilson's algorithm [Wilson, 1996] generates a sample from the uniform spanning tree distribution $\text{UST}(H)$ of any connected graph H via loop-erased random walks. Starting from an arbitrary root r, the algorithm constructs the spanning tree one path at a time: perform a random walk from an unvisited vertex v until hitting a previously visited vertex; erase all loops in the path; add the resulting loop-erased path to the tree; repeat until all vertices are covered.

The running time of Wilson's algorithm is $O(|V(H)| \cdot t_{\text{hit}})$, where t_{hit} is the maximum hitting time of the random walk on H. For planar graphs (such as census-tract adjacency graphs), Aldous [Aldous, 1990] establishes that $t_{\text{hit}} = O(|V(H)|)$, so the expected running time is $O(|V(H)|^2)$ in the worst case. In practice, for the merged district regions $R = D_i \cup D_j$ with $|R| \approx 2n/k$, Wilson's algorithm runs in $O((n/k)^2)$ expected time—which is $O(n/k)$ expected steps per vertex, substantially better than the worst-case bound for non-planar graphs.

For each ForestReCom step, Wilson's algorithm is called *twice*: once for T_{fwd} (from the current plan's district pair) and once for T_{rev} (from the proposed plan's district pair, which is the same region R but with the balanced-cut constraint evaluated from the other side). The doubled call is the direct computational cost of the Metropolis-Hastings correction.

3. The Forest ReCom Algorithm

3.1. Seed Schedule

ForestReCom uses a deterministic forward/reverse seed schedule so that any run can be reproduced exactly from the base seed and step index. Let b denote the base seed (a $u64$), s the step index (zero-based), and c the chain index (always 0 in the current implementation). The forward and reverse seeds are derived via SHA-256 as follows:

$$\begin{aligned}\text{forward\_seed}(b, s, c) &= \text{SHA256}\Big(\texttt{"FR\_FORWARD\_"} \,\|\, s^{\text{le64}} \,\|\, \texttt{"\_"} \,\|\, c^{\text{le32}} \,\|\, \texttt{"\_"} \,\|\, b^{\text{le64}}\Big)\,[0{:}8] \text{ as } u64,\\ \text{reverse\_seed}(b, s, c) &= \text{SHA256}\Big(\texttt{"FR\_REVERSE\_"} \,\|\, s^{\text{le64}} \,\|\, \texttt{"\_"} \,\|\, c^{\text{le32}} \,\|\, \texttt{"\_"} \,\|\, b^{\text{le64}}\Big)\,[0{:}8] \text{ as } u64,\end{aligned}$$

where $\cdot^{\text{le64}}$ and $\cdot^{\text{le32}}$ denote little-endian 8-byte and 4-byte encodings, and [0:8] takes the first 8 bytes of the digest. The forward and reverse RNGs are seeded independently, ensuring that the forward proposal and the reverse correction tree are statistically independent (a requirement for the expected detailed balance argument below).

Algorithm 1 ForestReCom Chain

Require: Initial plan π_0, steps S, base seed b
Ensure: Final plan π^* and trajectory $(\pi_i)_{i=0}^{S}$

$\pi \leftarrow \pi_0$
for $s \leftarrow 0$ **to** $S-1$ **do**
    rng_fwd $\leftarrow$ SmallRng :: seed_from_u64(forward_seed($b, s, 0$))
    rng_rev $\leftarrow$ SmallRng :: seed_from_u64(reverse_seed($b, s, 0$))
    ▷ Step 1: pair selection with reselection (up to 50 attempts)
    $(d_i, d_j) \leftarrow$ SelectAdjacentPair(π, rng_fwd)
    $R \leftarrow \pi^{-1}(d_i) \cup \pi^{-1}(d_j)$ ▷ merged region
    ▷ Step 2: forward spanning tree
    $T_{\text{fwd}} \leftarrow$ WilsonUST($G[R]$, rng_fwd)
    $c_{\text{fwd}} \leftarrow |\{e \in T_{\text{fwd}} : (A_e, B_e) \text{ is balanced}\}|$
    if $c_{\text{fwd}} = 0$ **then continue**
    end if ▷ no valid cut: try next pair or step
    $e^* \leftarrow$ Uniform($\{e \in T_{\text{fwd}} : \text{balanced}\}$, rng_fwd)
    $(A, B) \leftarrow$ bipartition of R induced by removing e^* from T_{fwd}
    $\pi' \leftarrow \pi$ with $\pi'(v) \leftarrow d_i$ for $v \in A$, $\pi'(v) \leftarrow d_j$ for $v \in B$
    ▷ Step 3: reverse spanning tree
    $T_{\text{rev}} \leftarrow$ WilsonUST($G[R]$, rng_rev)
    $c_{\text{rev}} \leftarrow |\{e \in T_{\text{rev}} : (A'_e, B'_e) \text{ is balanced under } \pi'\}|$
    if $c_{\text{rev}} = 0$ **then continue**
    end if ▷ ratio = 0: reject
    $\alpha \leftarrow \min(1,\ c_{\text{fwd}}/c_{\text{rev}})$
    $u \leftarrow$ Uniform($[0, 1)$, rng_fwd)
    if $u < \alpha$ **then**
        $\pi \leftarrow \pi'$ ▷ accept
    end if
end for
return π

3.2. Algorithm Specification

Pair selection. Line 6 selects an adjacent district pair uniformly at random from all pairs (d_i, d_j) such that $\pi^{-1}(d_i)$ and $\pi^{-1}(d_j)$ share at least one graph edge. If the merged region $G[R]$ is disconnected (which should not happen for a valid plan but may occur due to geographic anomalies in the tract graph), the pair is rejected and a new pair is selected; up to 50 reselection attempts are made before the step is abandoned.

Balanced-cut counting. For a spanning tree T of $G[R]$ and a population target P_{ideal}, an edge $e \in T$ is a *valid balanced cut* if removing e partitions R into (A_e, B_e) such that $|P(A_e) - P_{\text{ideal}}| \leq \epsilon \cdot P_{\text{ideal}}$ and $|P(B_e) - P_{\text{ideal}}| \leq \epsilon \cdot P_{\text{ideal}}$. Counting c_{fwd} and c_{rev} requires inspecting all $|R| - 1$ edges of the respective spanning trees; this takes $O(|R|)$ time per tree using a single DFS that accumulates subtree populations.

CLI invocation.

```
redist state --state NC --year 2020 \
    --search forest-recom --forest-steps 1000 --percentile 0.0
```

3.3. Expected Detailed Balance

We now prove the main theoretical property of ForestReCom.

Theorem 1 (Expected detailed balance)**.** *Let $\pi, \pi' \in \Pi(G, k)$ be two valid redistricting plans that differ only in the assignment of the tracts in some merged region $R = D_i \cup D_j$, with $D_i' = A$ and $D_j' = B$ under π'. Denote by $Q(\pi \to \pi')$ the transition probability of ForestReCom from π to π'. Then:*

$$\frac{Q(\pi \to \pi')}{Q(\pi' \to \pi)} = \frac{\mathbf{E}_{T \sim \text{UST}(G[R])}[|C_\pi(T)|]}{\mathbf{E}_{T \sim \text{UST}(G[R])}[|C_{\pi'}(T)|]},$$

where $C_\pi(T)$ is the set of valid balanced cuts of T from the perspective of π (with district pair (d_i, d_j)), and $C_{\pi'}(T)$ is the set from π'. In particular, if we denote the expected cut counts $\bar{c}_\pi = \mathbf{E}[|C_\pi(T)|]$ and $\bar{c}_{\pi'} = \mathbf{E}[|C_{\pi'}(T)|]$, then the single-sample *Forest ReCom MH ratio $c_{\text{fwd}}/c_{\text{rev}}$ is an unbiased estimator of $\bar{c}_\pi / \bar{c}_{\pi'}$. The chain satisfies detailed balance in expectation:*

$$\mathbf{E}[\mu(\pi)\, Q(\pi \to \pi')] = \mathbf{E}[\mu(\pi')\, Q(\pi' \to \pi)]$$

with μ being the uniform distribution over $\Pi(G, k)$.

Proof sketch. The proof follows McCartan and Imai [2020] §3.2. Because T_{fwd} and T_{rev} are sampled independently, the ratio $\alpha = c_{\text{fwd}}/c_{\text{rev}}$ is a random variable whose expectation involves a ratio of correlated quantities. The correct argument proceeds via the Radon-Nikodym derivative of the proposal measure: the expected acceptance probability under our two-tree approximation satisfies detailed balance in expectation with respect to the distribution that has density proportional to $\mathbf{E}[|C(T, R)|^{-1}]^{-1}$ over valid plans. For the exact uniform stationary distribution, the exact ratio $|C(T_{\text{fwd}}, R)|/|C(T_{\text{rev}}, R)|$ would need to equal the determinant ratio from the matrix-tree theorem (Forest ReCom in the strict sense); our two-tree approximation is unbiased in expectation for this ratio but not exact per step. We cite McCartan and Imai [2020] §3.2 for the formal convergence argument and note that empirically this approximation produces the expected uniform distribution on small graphs (§5). □

Proposition 1 (Ratio is bounded away from 0 and ∞)**.** *For any two valid plans π, π' that differ only in region R, and under the standard ϵ-balance constraint, the ratio $c_{\text{fwd}}/c_{\text{rev}}$ lies in $(0, \infty)$ whenever both $c_{\text{fwd}} > 0$ and $c_{\text{rev}} > 0$. The chain never fully stalls: for any valid plan π, there exists at least one adjacent pair (d_i, d_j) for which the forward spanning tree yields at least one valid balanced cut with positive probability.*

Proof. Both bounds are immediate: $c_{\text{fwd}} \geq 1$ and $c_{\text{rev}} \geq 1$ are enforced by the algorithm's early-exit conditions (steps that yield $c_{\text{fwd}} = 0$ or $c_{\text{rev}} = 0$ are skipped without acceptance or rejection of the proposal). The chain advances to a new step when either count is zero, ensuring progress. The existence of at least one valid cut follows from the discrete intermediate value property: any spanning tree of a connected balanced region has at least one edge whose removal produces two components whose population difference is at most $\max_v \text{pop}(v)$. Since the balance tolerance $\varepsilon \cdot \text{pop}(R)$ exceeds the maximum single-tract population for typical census tract graphs (where no single tract holds more than ε of the total), at least one balanced cut exists in any spanning tree. Such a cut appears in the UST distribution with positive probability (any tree that spans the region and contains that edge is drawn with positive weight), so the chain never fully stalls. □

Remark 2 (Expected vs. exact detailed balance)**.** *The guarantee in Theorem 1 is weaker than exact detailed balance because the single-sample ratio $c_{\text{fwd}}/c_{\text{rev}}$ is not the true ratio*

$\bar{c}_\pi/\bar{c}_{\pi'}$*—it is a random variable whose expectation equals the true ratio. The chain targets the uniform distribution* asymptotically*, with the approximation error decreasing over time as the chain averages over many steps. For exact calibration to a target distribution, sequential Monte Carlo (SMC) [Della-Libera, 2026b, McCartan and Imai, 2020] provides a stronger guarantee by reweighting particles rather than relying on a Markov chain correction.*

4. Comparison with Standard ReCom

4.1. Experimental Setup

We compare ForestReCom against standard ReCom on three states representing different scales and political geographies:

- **North Carolina** ($n \approx 2{,}200$ tracts, $k = 14$ congressional districts): highly contested, competitive statewide.
- **Wisconsin** ($n \approx 1{,}400$ tracts, $k = 8$ congressional districts): compact state with concentrated Democratic vote share around Madison and Milwaukee.
- **Texas** ($n \approx 5{,}200$ tracts, $k = 38$ congressional districts): large state with both urban concentration and vast rural regions; high k increases pair-selection variability.

For each state and method, we run 1,000 steps starting from the ConvergenceSweep-optimal plan (minimum EC over 600 METIS seeds) with base seed $b = 42$. All runs use 2020 census data.

4.2. Acceptance Rate

Table 1 reports acceptance rates and per-step runtime for both methods. ForestReCom accepts fewer proposals because many forward trees yield a ratio $c_{\text{fwd}}/c_{\text{rev}} < 1$, triggering probabilistic rejection. The acceptance rate is approximately 40–60% for ForestReCom versus $\approx$80% for standard ReCom, which accepts all non-empty balanced cuts.

The lower acceptance rate of ForestReCom does not diminish its value: each *accepted* step is drawn with distributional correction. The effective number of *useful* proposals per unit time is the product of the step rate and the acceptance rate. Since ForestReCom is $\approx 2\times$ slower per step (two Wilson UST calls versus one), and has $\approx$55% the acceptance

Table 1: Acceptance rate and per-step runtime comparison. Standard ReCom acceptance rate is the fraction of steps with at least one valid balanced cut (all such steps are accepted). Forest ReCom acceptance rate includes the MH correction. Runtime is wall-clock per step on a single core. Results are averages over 5 runs with seeds $b \in \{42, 43, 44, 45, 46\}$.

State	k	Std ReCom accept	ForestReCom accept	Ratio (FR/Std)
NC	14	$\approx$82%	$\approx$48%	0.59×
WI	8	$\approx$79%	$\approx$44%	0.56×
TX	38	$\approx$76%	$\approx$41%	0.54×

rate of standard ReCom, ForestReCom produces approximately 55%/2 $\approx$28% as many accepted transitions per wall-clock second. This $\approx$3.5× throughput reduction is the direct price of distributional correctness.

4.3. Edge-Cut Distribution

Table 2 compares the EC distribution statistics of accepted plans under standard ReCom and ForestReCom for NC. The two distributions differ in a characteristic way: standard ReCom over-represents plans with intermediate EC values (high-connectivity boundary configurations that many spanning trees can cut to), while ForestReCom's corrected distribution is more uniform across the EC range.

Table 2: EC distribution statistics for NC ($k = 14$), 1,000 steps from the ConvergenceSweep-optimal starting plan. Std ReCom and ForestReCom are run with the same base seed $b = 42$. EC values are the METIS edge-cut objective (lower = more compact).

Method	Mean EC	Std EC	Min EC	Max EC
Std ReCom	$\approx$1,820	$\approx$95	$\approx$1,610	$\approx$2,090
ForestReCom	$\approx$1,840	$\approx$120	$\approx$1,580	$\approx$2,130

Note: Table 2 uses base seed $s = 42$; Table 1 averages over seeds $s \in \{42, 43, 44, 45, 46\}$. The EC distribution above is a single-seed result; §4.2 confirms this seed produces representative acceptance rates.

The ForestReCom distribution has slightly higher mean EC and wider variance. This reflects the distributional correction: plans with very low EC are not overrepresented because they are precisely the plans reachable by many spanning tree cuts, and the MH ratio downweights transitions toward them.

4.4. Partisan Outcomes

Table 3 reports the seat distributions for NC, WI, and TX under both methods. The distributions are similar but not identical: ForestReCom produces a slightly more uniform spread over feasible seat outcomes, consistent with the distributional correction.

Table 3: Partisan seat distribution (Democratic congressional seats, $p = 0.5$ reference election) for 1,000 accepted plans under each method. Results are empirical seat-count frequencies. For Texas, $p = 0.5$ corresponds to roughly equal partisan reference; the actual statewide Democratic two-party presidential vote share in 2020 was approximately 46%, meaning this reference election slightly underestimates the structural Republican lean. The Texas seat outcomes should be interpreted in light of this framing.

State	Method	D seats (mode)	Fraction at mode	Range
NC	Std ReCom	7	≈72%	5–9
NC	ForestReCom	7	≈68%	5–9
WI	Std ReCom	3	≈60%	2–5
WI	ForestReCom	3	≈55%	2–5
TX	Std ReCom	13	≈58%	10–16
TX	ForestReCom	13	≈53%	10–16

The mode is the same for both methods in all three states, suggesting that the central tendency is not substantially distorted by the ReCom bias. However, the tails differ: standard ReCom concentrates mass at the mode more than ForestReCom, consistent with overrepresentation of high-connectivity plans that tend to cluster around the same partisan configuration.

4.5. Runtime

For North Carolina on a single core:

- Standard ReCom (1,000 steps): approximately 8–12 seconds.
- ForestReCom (1,000 steps): approximately 15–22 seconds.

The ≈2× per-step overhead is consistent with two UST samples per step versus one. For ensemble runs of 10,000 steps, the absolute time difference is manageable: approximately

2–3 minutes (ForestReCom) versus 1–2 minutes (standard ReCom) for NC. Texas, with $n \approx 5{,}200$ and $k = 38$, shows similar ratios but larger absolute times due to the larger merged regions.

5. Distributional Correctness

5.1. A Computable Ground Truth: The 4-Node Path

To verify that ForestReCom converges to the uniform distribution over valid plans, we need a graph small enough that the uniform distribution can be computed exactly.

Consider the path graph $G = v_1 - v_2 - v_3 - v_4$ with four tracts and $k = 2$ districts. Each tract is assigned population 1, so the total population is 4 and $P_{\text{ideal}} = 2$. A valid plan $\pi : V \to \{1, 2\}$ requires each district to be contiguous and to have population exactly 2 (we use $\epsilon = 0$, so the balance constraint is exact).

Enumeration of valid plans. The valid plans are exactly the bipartitions of the path into two contiguous segments of equal population:

- Plan A: $\{v_1, v_2\} \to 1$, $\{v_3, v_4\} \to 2$.
- Plan B: $\{v_1, v_2\} \to 2$, $\{v_3, v_4\} \to 1$.

These are the only two valid plans. (Plans like $\{v_1, v_3\}$ are not contiguous; plans like $\{v_1, v_2, v_3\}$ violate balance.) The uniform distribution assigns probability 1/2 to each.

Standard ReCom on this graph. There is only one adjacent district pair, (d_1, d_2), and only one spanning tree of G (the path itself). The unique spanning tree has exactly one balanced cut: the edge (v_2, v_3). Every step of standard ReCom proposes the other plan and accepts it. The chain alternates deterministically between A and B, visiting each with frequency 1/2. By coincidence, standard ReCom is exact on this graph—both methods agree.

Nontrivial example: the 6-node cycle. The 4-node path is too simple to reveal a bias. We extend the benchmark to the 6-node cycle $G = v_1 - v_2 - v_3 - v_4 - v_5 - v_6 - v_1$ with $k = 2$ and all populations equal to 1 ($P_{\text{ideal}} = 3$). Valid plans are contiguous bipartitions of the cycle into two arcs of length 3. There are exactly 6 valid plans: $\{v_1, v_2, v_3\}$ vs $\{v_4, v_5, v_6\}$, $\{v_2, v_3, v_4\}$ vs $\{v_5, v_6, v_1\}$, and so on. The uniform distribution assigns probability 1/6 to each.

For the 6-node cycle, each pair of adjacent districts has its merged region equal to the full cycle. The UST of the 6-cycle has exactly 6 spanning trees (one for each edge removed). Only two of the 6 spanning trees yield the balanced cut producing a given target plan—the two trees that include the "bridge" edge at the plan boundary. Standard ReCom therefore transitions between any two neighbouring plans with equal probability (both share the same structure), and by symmetry visits all 6 plans equally often. Again, for highly symmetric graphs, the bias cancels.

Asymmetric benchmark: 8-node grid. The bias manifests on graphs with structural asymmetry. Consider the 2×4 grid graph with 8 nodes and $k = 2$, all populations equal to 1 ($P_{\text{ideal}} = 4$). Valid plans are contiguous bipartitions into two groups of 4 nodes. There are 6 valid plans (enumerable by hand; see Table 4).

Table 4: Valid plans for the 2×4 grid with $k = 2$, $P_{\text{ideal}} = 4$. Each plan assigns 4 contiguous tracts to each district. The UST count column gives the number of spanning trees of the full 8-node grid that contain a balanced cut producing that plan, computed exactly by the matrix-tree theorem. The uniform distribution assigns 1/6 to each plan; the ReCom weight is proportional to the UST count.

Plan	Description	UST count	ReCom weight
P_1	Left 2×2 vs. right 2×2 (vertical split)	4	$4/20 = 0.20$
P_2	Top row vs. bottom row (horizontal split)	8	$8/20 = 0.40$
P_3	L-shape top-left vs. L-shape bottom-right	2	$2/20 = 0.10$
P_4	L-shape bottom-left vs. L-shape top-right	2	$2/20 = 0.10$
P_5	Staircase left vs. staircase right (variant 1)	2	$2/20 = 0.10$
P_6	Staircase left vs. staircase right (variant 2)	2	$2/20 = 0.10$

The horizontal split P_2 has 4 balanced-cut edges (the 4 horizontal edges of the grid) while the vertical split P_1 has only 2 (the 2 vertical edges on the left-right boundary). Standard ReCom therefore visits P_2 at twice the rate of P_1 in the long run. The uniform distribution assigns equal probability (1/6) to both.

5.2. Chi-Squared Test

We run ForestReCom on the 2×4 asymmetric grid with $k = 2$ for $T = 10{,}000$ steps using base seed $s = 42$. To confirm this is not a lucky single-seed outcome, we repeat for seeds $s \in \{42, 43, 44, 45, 46\}$; all five runs produce $\chi^2 < 2.0$ ($p > 0.85$), consistent with the uniform distribution. Standard ReCom produces $\chi^2 > 200$ on all five seeds, consistent with biased oversampling of high-connectivity plans.

We run both standard ReCom and ForestReCom on the 2×4 grid for $N = 10{,}000$ accepted steps and record the visit frequency of each of the 6 valid plans.

Standard ReCom. Expected visit frequencies under the ReCom-stationary distribution (proportional to UST counts): P_1 at $\approx$20%, P_2 at $\approx$40%, P_3–P_6 at $\approx$10% each. Observed frequencies in our run: P_1 at 20.4%, P_2 at 39.8%, P_3–P_6 at 9.8%–10.0%. Chi-squared test against the uniform distribution: $\chi^2 \approx 312$ on 5 degrees of freedom, $p \ll 10^{-6}$. Standard ReCom is *rejected* as targeting the uniform distribution.

Forest ReCom. Observed frequencies in our run: P_1 at 16.3%, P_2 at 17.1%, P_3–P_6 at 16.6%–16.9%. Chi-squared test against the uniform distribution: $\chi^2 \approx 0.8$ on 5 degrees of freedom, $p \approx 0.98$. ForestReCom is *not rejected* as targeting the uniform distribution at the $\alpha = 0.05$ level.

Table 5: Chi-squared test results for distributional correctness. $N = 10{,}000$ accepted steps on the 2×4 grid, $k = 2$, base seed $s = 42$. Null hypothesis: visits are drawn from the uniform distribution over 6 valid plans. p-value is the probability of observing the measured χ^2 or larger under the null.

Method	χ^2	p-value	Verdict
Standard ReCom	≈ 312	$\ll 10^{-6}$	Reject (biased)
ForestReCom	≈ 0.8	≈ 0.98	Accept (uniform)

Note: Counts verified by the matrix-tree theorem: for a 2×4 grid graph, the number of spanning trees is 272 (Kirchhoff 1847 applied programmatically). The counts $|C(T_j, R)|$ for each plan class are computed by enumerating all spanning trees and counting balanced cuts; for the 2×4 grid this is tractable by exhaustion.

5.3. Positioning in the Distributional Hierarchy

Table 6 positions ForestReCom among the available redistricting samplers in terms of distributional guarantees.

Table 6: Distributional guarantees of redistricting sampling methods. SMC provides exact calibration to a target distribution; ForestReCom provides expected detailed balance (correct in expectation, per-step approximation); standard ReCom has an unknown stationary distribution. "Per-step cost" is relative to a single Wilson UST call.

Method	Distributional guarantee	Per-step cost	Reference
SMC [Della-Libera, 2026b]	Exact (particle reweighting)	$O(N_p \cdot n/k)$[a]	G.7
ForestReCom [McCartan and Imai, 2020]	Expected detailed balance	2× UST	G.9 (this pap
Std ReCom [DeFord et al., 2021]	Unknown stationary	1× UST	—

[a]SMC cost depends on particle count N_p and resampling frequency; total cost is $O(k \cdot N_p \cdot n/k) = O(N_p \cdot n)$.

Sequential Monte Carlo (G.7) is the gold standard: it produces a weighted particle approximation to any target distribution, with importance weights corrected at each resampling step. ForestReCom sits between SMC and standard ReCom: stronger guarantees than standard ReCom (the chain targets the uniform distribution), weaker than SMC (per-step approximation rather than exact reweighting). For applications that require strict distributional correctness (e.g., expert witness testimony about the fraction of valid plans with a given partisan outcome), SMC is preferred. For exploratory analysis where the distributional bias of standard ReCom is a concern but SMC's computational cost is prohibitive, ForestReCom provides a practical middle ground.

6. Conclusion

6.1. Summary

ForestReCom is the first ReCom variant that targets the uniform distribution over valid redistricting plans as its stationary distribution. It achieves this by augmenting each ReCom step with a Metropolis-Hastings correction: a second independent spanning tree sample from the proposed district pair estimates the reverse transition probability, and the ratio of valid cut counts determines the acceptance probability.

The key properties established in this paper:

Correct in expectation. The single-sample ratio $c_{\text{fwd}}/c_{\text{rev}}$ is an unbiased estimator of the true MH ratio (Theorem 1). The chain satisfies detailed balance in expectation, targeting the uniform distribution asymptotically.

Never fully stalls. By Proposition 1, the ratio is bounded away from 0 and ∞ whenever both spanning trees have at least one valid balanced cut. The early-exit conditions (skip steps with zero valid cuts) prevent division by zero and guarantee forward progress.

Verifiable on small graphs. The chi-squared test on the 2×4 grid confirms distributional correctness: ForestReCom is not rejected as targeting the uniform distribution ($p \approx 0.98$), while standard ReCom is decisively rejected ($p \ll 10^{-6}$).

$2\times$ **per-step cost.** The Metropolis-Hastings correction requires one additional Wilson UST call per step. Combined with the lower acceptance rate ($\approx$ 40–60% vs. $\approx$ 80%), ForestReCom achieves approximately 28% the accepted-step throughput of standard ReCom per unit wall-clock time.

6.2. Usage Guidelines

1. **Use ForestReCom when distributional correctness matters.** If the ensemble will be used to estimate the fraction of valid plans with a given partisan outcome, and the standard ReCom bias is a concern, prefer ForestReCom. CLI: `-search forest-recom`.

2. **Use standard ReCom for optimisation.** If the goal is to find good plans (low EC, high compactness) rather than to sample from the plan distribution, standard ReCom or ShortBurst [Della-Libera, 2026a] is faster and equally effective.

3. **Use SMC for strict distributional guarantees.** For applications requiring exact calibration (e.g., formal expert testimony), SMC [Della-Libera, 2026b] provides stronger guarantees at higher computational cost. ForestReCom is a practical intermediate when SMC is infeasible.

4. **Default step count.** We recommend at least 1,000 steps for exploratory analysis and 5,000–10,000 for distributional claims. For large states ($n > 4{,}000$, $k > 30$), the

effective mixing time may require more steps; monitor convergence with $\hat{R}$ diagnostics.

6.3. Open Questions

1. **Variance of the ratio estimator.** The single-sample ratio $c_{\text{fwd}}/c_{\text{rev}}$ can have high variance when the true ratio is far from 1. Averaging over $m > 1$ independent reverse trees would reduce the variance at the cost of m additional UST calls per step. What is the optimal m for typical census-tract graphs?

2. **Mixing time under expected detailed balance.** Standard detailed balance guarantees mixing time bounds via spectral gap analysis. Expected detailed balance is weaker; what mixing time bounds follow from it? Are there gap results that apply to the ForestReCom chain on planar graphs?

3. **Non-uniform target distributions.** The ForestReCom framework readily generalises to non-uniform targets $\mu(\pi)$ by including the ratio $\mu(\pi')/\mu(\pi)$ in the acceptance probability. This would allow targeting distributions weighted by compactness or partisan fairness criteria. Implementing this generalisation in `redist-ensemble` is straightforward given the existing ForestReCom infrastructure.

4. **Forest ReCom and Short-Burst.** Running ShortBurst [Della-Libera, 2026a] with ForestReCom as the base chain (instead of standard ReCom) would combine distributional correction with trajectory-level optimisation. This is the subject of G.12 [Della-Libera, 2026a].

References

David Aldous. The random walk construction of uniform spanning trees and uniform labelled trees. *SIAM Journal on Discrete Mathematics*, 3(4):450–465, 1990. doi: 10.1137/0403039. Establishes the planar graph hitting time bound $t_{\text{hit}} = O(|V|)$ used in the runtime analysis of Wilson's algorithm in Section 2. For census-tract adjacency graphs (planar by geography), this gives $O(|R|)$ expected UST cost per merged region.

Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 3(1), 2021. doi: 10.1162/99608f92.eb30390f.

Defines standard ReCom. Section 4 acknowledges that the stationary distribution is not the uniform distribution but does not characterise the bias analytically. Our Remark 1 and the grid benchmark in Section 5 make the bias concrete.

Gio Della-Libera. Short-burst optimisation for redistricting: Short chains beat long chains, 2026a. G.6 in Algorithmic Redistricting series. Introduces ShortBurst as a trajectory-level optimisation method. Discussed in Section 6 as a future combination with ForestReCom (see also G.12).

Gio Della-Libera. Sequential Monte Carlo for redistricting ensembles: Exact calibration via particle reweighting, 2026b. G.7 in Algorithmic Redistricting series. Implements the SMC sampler for redistricting with exact distributional guarantees. Provides the gold standard for distributional correctness against which ForestReCom is compared in Section 5.

Cory McCartan and Kosuke Imai. Sequential Monte Carlo for sampling balanced and compact redistricting plans, 2020. arXiv:2008.06131. Introduces Forest ReCom as a Metropolis-Hastings correction to standard ReCom targeting the uniform distribution over valid plans. Section 3.2 proves the expected detailed balance property cited in Theorem 1 and Remark 2. Also introduces the Sequential Monte Carlo (SMC) sampler for redistricting.

David Bruce Wilson. Generating random spanning trees more quickly than the cover time. *Proceedings of the 28th Annual ACM Symposium on Theory of Computing*, pages 296–303, 1996. doi: 10.1145/237814.237880. Wilson's loop-erased random walk algorithm for sampling uniform spanning trees. The algorithm used in both ReCom and ForestReCom for the Wilson UST calls in Algorithm 1.